

Thyroid Disorders 2: Hypothyroidism, Thyroiditis, Thyroid Carcinoma & Solitary Thyroid Nodule

Internal Medicine Department | Source: 2-Thyroid - Hypothyroidism & Thyroiditis, Thyroid Carcinoma, Solitary Thyroid Nodule.pdf (45 slides)

Legend: ★ = high-yield / likely exam target · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

⚠ The two thyroid-cancer slides give **different frequencies** (table vs infographic). Both are shown and flagged in §8.

🔗 **Bridge from Thyroid Lecture 1:** Lecture 1 covered the axis (TRH → TSH → T4/T3), the tests (TSH first; primary disease moves TSH and T4 in opposite directions) and the **too-much-hormone** states. This lecture is the mirror image: **too little hormone**, the **inflammatory** conditions that can swing either way, and **structural** disease (nodules and cancer). It ends with lab-based algorithms that tie both lectures together.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Pathophysiology & aetiology of hypothyroidism and other thyroid disorders	Goitrous vs non-goitrous causes
Knowledge	★ Manifestations, complications, management of hypothyroidism	Carpal tunnel, anaemia types, myxoedema coma
Skill	★ Interpret clinical findings & investigations	TSH algorithms, subclinical disease
Skill	★ Management plans (thyroxine dosing, thyroiditis, cancer, nodule work-up)	Slow vs rapid replacement; FNA indications
Skill	★ Emergencies	Myxoedema coma treatment

Lecture flow: Hypothyroidism (causes → features by system → myxoedema coma → investigations/DDx → treatment) → Thyroiditis (subacute, Hashimoto) → Solitary thyroid nodule → Thyroid cancer → Subclinical hypo/hyperthyroidism → Diagnostic algorithms & thyrotoxicosis table → Case.

2. Main Content (Lecture Order)

2.1 Hypothyroidism: Causes (Slides 4-7)

Level	Share
★ Primary (thyroid)	★ 95%
Secondary (hypothalamic/pituitary)	5%
Tissue resistance to thyroid hormone	Rare

★★ Primary hypothyroidism: non-goitrous vs goitrous (Slide 6):

Non-goitrous (small/absent gland)	Goitrous (enlarged gland)
Congenital developmental defect	★ Pendred syndrome : inherited biosynthetic defect + congenital deafness
★ Post-surgery	★ Iodine deficiency → endemic goitre
★ Post-I-131 therapy	Maternally transmitted (antithyroid drug or I-131 in pregnancy)
Idiopathic (atrophic) : may have antithyroid or TSH-receptor-blocking antibodies; associated with other endocrine autoantibodies and autoimmune disease (SLE, myasthenia gravis)	Drug-induced : phenylbutazone, PAS , antithyroid drugs
	★ Chronic thyroiditis: Hashimoto's, Riedel's
	Self-limited: subacute thyroiditis, postpartum thyroiditis

Myxoedema vs hypothyroidism (Slide 7): myxoedema = skin and tissue swelling from **accumulated mucopolysaccharides**; it is a **sign**, not a disease. Hypothyroidism is the broader condition. ★ **Epidemiology**: much more common in ♀ **than** ♂, usually **30-50 y**, mostly **autoimmune** (thyroid antibodies in > 80%).

🔄 **Recap: Causes.** 95% primary. Non-goitrous = gland removed, ablated or absent. Goitrous = gland trying (iodine deficiency, Pendred, drugs) or inflamed (Hashimoto, subacute, postpartum). Mostly autoimmune women aged 30-50.

🔗 **Link:** Whatever the cause, low hormone slows every system. The lecture walks through them one by one.

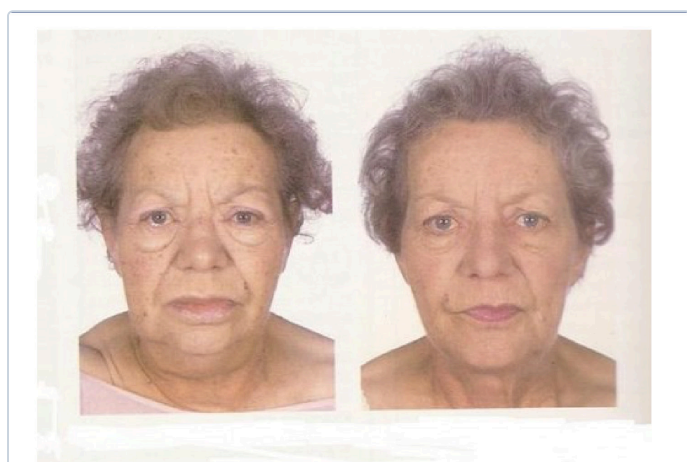
📌 **Up next: Clinical picture by system.**

2.2 Hypothyroidism: Clinical Picture (Slides 8-14)

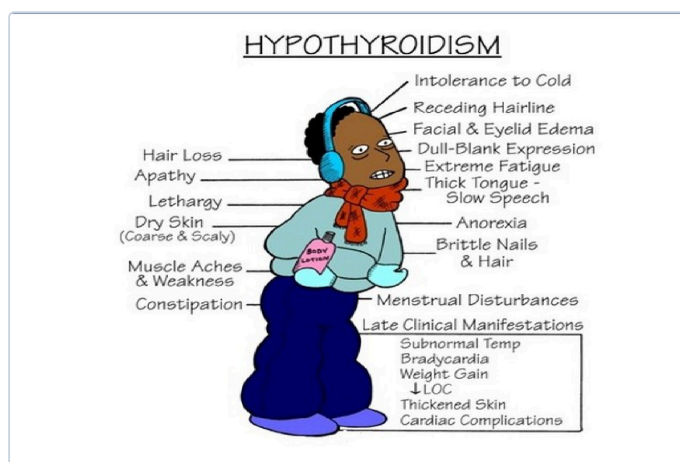
#	System	Features
1	★ Neurological	↓ memory, mental slowing, dementia (" myxoedema madness "), depression · ★ delayed relaxation of tendon jerks ("hung-up" reflexes) · Mucinous infiltration : flexor retinaculum → ★ carpal tunnel syndrome ; inner ear → progressive deafness ; vocal cords → hoarseness ; tongue → slurred speech
2	★ Cardiovascular	★ Sinus bradycardia · cardiomyopathy → HF · cholesterol pericarditis / pericardial effusion · atherosclerosis → angina, claudication · ★ hypertension (↑ peripheral resistance, diastolic [added])
	★ Haematological (anaemia)	Normocytic : marrow depression & ↓ O ₂ need · Megaloblastic : associated pernicious anaemia · Microcytic : menorrhagia & achlorhydria
3	Pulmonary	Pleural effusion · ↓ ventilatory response to hypoxia/hypercapnia → CO₂ retention
4	Renal	↓ free-water excretion → ★ hyponatraemia (may be SIADH)
5	GIT	Adynamic ileus : constipation, even obstruction · achlorhydria (pernicious anaemia) · ascites (high cholesterol)
6	Musculoskeletal	Arthralgia, joint effusions, stiff muscles

#	System	Features
7	★ Skin & hair	Puffy face, coarse features · dry cold skin · ★ orange tint (carotenaemia: carotene → vitamin A conversion needs thyroxine) · malar flush · hair sparse, brittle, coarse · ★ loss of the outer 1/3 of eyebrows · xanthelasma (↑ cholesterol)
8	Reproductive	★ Menorrhagia (anovulatory cycles): the commonest · late: amenorrhoea & galactorrhoea (↑ prolactin)
9	Metabolic / endocrine	Growth retardation in children (juvenile myxoedema) · GH deficiency · ★ weight gain despite ↓ appetite · hypothermia, cold intolerance · hyperlipidaemia (↓ lipoprotein degradation)
10	Thyroid gland	Atrophic, enlarged, or post-thyroidectomy scar

Mnemonic [added]: **"Everything slows, swells, and gets cold"**: slow brain/heart/gut/reflexes; mucin swelling (carpal tunnel, voice, face); cold, dry, pale-orange skin.



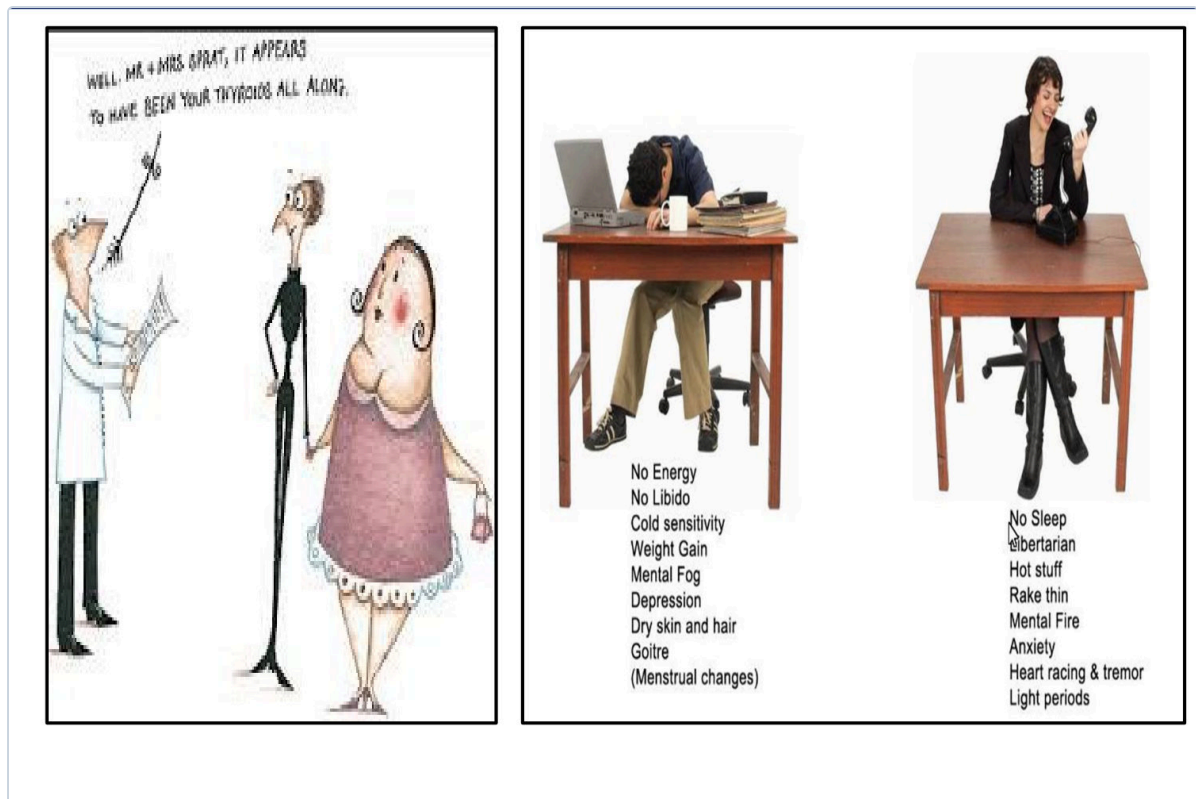
Slide 12: Facial features of hypothyroidism before and after treatment



Slide 14: Hypothyroidism features overview

Findings	Hypothyroidism	Hyperthyroidism
Metabolic	Intolerance to cold Weight gain Decreased appetite	Intolerance to heat Weight loss Increased appetite
Neuropsychiatric	Excessive fatigue	Irritability, restlessness
Periorbital findings	Periorbital edema (due to generalized myxedema)	Periorbital edema, lid lag, and exophthalmos (in Graves opthalmopathy)
Cardiovascular	Bradycardia, decreased cardiac output	Tachycardia, palpitations, hypertension
Skin	Cold, dry	Warm, moist
Gastrointestinal	Constipation	Hyperdefecation
Musculoskeletal	Cramps	Osteopathy
Edema	Generalized myxedema (initially pretibial)	Pretibial myxedema (in Graves disease)
Myopathy	Proximal	
Reproduction	♀ Menstrual disorders ♂ Decreased libido, infertility	
Hair	Hair loss	

Slide 20: Hypothyroidism vs hyperthyroidism, system by system



Slide 19: Hypo vs hyper, cartoon comparison

★★★★ Myxoedema coma (Slide 15):

- **Causes:** long-standing **untreated** hypothyroidism; hypothyroid + exposure to ★ **cold, infection, trauma, CNS depressants**
- **Picture:** severe hypothyroid features + **bradycardia** · ★ **hypothermia** · ★ **dilutional hyponatraemia** · **alveolar hypoventilation → CO₂ retention & narcosis** · **loss of consciousness** · **hypoglycaemia** · **heart failure & hypotension**

🔄 **Recap: Clinical picture.** Slow brain and reflexes, carpal tunnel, deafness, hoarseness. Bradycardia, effusions, diastolic HTN. Three anaemia types. Hyponatraemia, constipation. Cold, dry, orange skin with lost outer eyebrows. Menorrhagia. Weight gain with poor appetite. Coma = hypothermia + hyponatraemia + hypoventilation + hypoglycaemia in an untreated patient after cold, infection or sedatives.

🔗 **Link:** Several conditions mimic this puffy, pale, high-cholesterol picture, so the lecture shows how to confirm the diagnosis and exclude look-alikes.

📌 **Up next: Investigations and differential diagnosis.**

2.3 Investigations & Differential Diagnosis (Slide 16)

Investigations:

1. ★ **Thyroid function** (TSH first, see algorithms in §2.8)
2. **CXR** → pericardial or pleural effusion
3. ★ **ECG** → **low voltage, bradycardia**

Differential diagnosis:

Condition	Similar to myxoedema	Different
Primary vs secondary hypothyroidism	Low T4	TSH ↑ (primary) vs ↓/normal (secondary)
★ Nephrotic syndrome	Facial puffiness, pallor · hypercholesterolaemia · ★ ↓ total T4/T3 (TBG lost in urine)	★ Normal TSH · heavy proteinuria
Chronic renal failure	Facial puffiness, pallor · hypertension · dry skin	Polyuria, pruritus · normal thyroid function

📌 **Recap: Diagnosis.** TSH + FT4, CXR for effusions, ECG with low voltage and bradycardia. Nephrotic syndrome lowers total T4 (lost TBG) but TSH stays normal. CRF looks similar but thyroid function is normal.

🔗 **Link:** With the diagnosis confirmed, treatment is simple in principle (replace T4), but speed matters: too fast can precipitate angina.

📌 **Up next: Treatment.**

2.4 Hypothyroidism: Treatment (Slides 17-18)

A. Preparations

- ★ **Levothyroxine (L-thyroxine): drug of choice**
- Desiccated thyroid extract (**T4 : T3 = 4 : 1**)
- Synthetic T3: **no real indication** (T4 is converted to T3 in the body)

B. ★ Rapid correction in:

- Neonatal, infantile, juvenile** hypothyroidism
- ★ **Myxoedema coma**
- Emergency surgery** in a hypothyroid patient → ★ **IV L-thyroxine + hydrocortisone**

C. ★ Slow correction in **elderly or cardiac patients**:

- Start ★ **25-50 µg/day** → ↑ by **25-50 µg every month** (clinical response) → full replacement ★ **150-200 µg/day**
- [added: fast replacement increases myocardial O₂ demand and can precipitate angina or MI]*

D. Clinical response: full effect takes ★ **2-3 months**.

E. ★ Monitoring (time to normalise):

Test	Normalises in
T4	A few days
T3 (better)	2-4 weeks
★ TSH	★ 6-8 weeks
<i>[added: so recheck TSH no sooner than 6-8 weeks after a dose change.]</i>	

★★★★ **Myxoedema coma treatment (Slide 18):**

- ★ **L-thyroxine 500 µg IV**
- ★ **Hydrocortisone 100 mg** (possible **adrenal insufficiency**)
- Hypertonic saline (+ glucose)** (hyponatraemia, hypoglycaemia)
- Assisted ventilation** (CO₂ retention)

5. ★ **Avoid further heat loss** [added: passive rewarming; active rewarming causes vasodilatation and collapse]

Mnemonic [added]: "**T-H-S-V-W**": **T**hyroxine, **H**ydrocortisone, **S**aline + sugar, **V**entilate, **W**arm passively. Give **steroid with (or before) thyroxine** [added].

🔁 **Recap: Treatment.** Levothyroxine. Rapid for neonates, coma and emergency surgery (IV + hydrocortisone). Slow in elderly/cardiac patients (25–50 µg, up 25–50 monthly to 150–200). Full effect 2–3 months; TSH takes 6–8 weeks. Coma: 500 µg IV T4 + hydrocortisone 100 mg + hypertonic saline/glucose + ventilation + no heat loss.

🔗 **Link:** Two causes of hypothyroidism listed earlier, subacute and Hashimoto's thyroiditis, can also cause temporary **thyrotoxicosis**. The lecture looks at thyroiditis next.

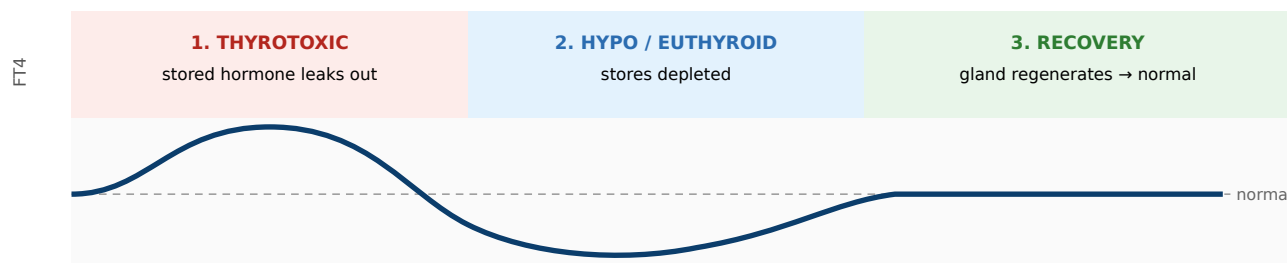
📺 **Up next: Thyroiditis.**

2.5 Thyroiditis (Slides 21–24)

A. ★★ Subacute (de Quervain) thyroiditis

- **Cause:** usually **follows an upper respiratory tract infection** (various **viruses**)
- **Clinical:** ★ **stretching of the capsule → pain over the thyroid**, or referred to the **lower jaw, ear, or occiput** · fever · nodularity · **dysphagia**
- **Investigations:** ★ **↑ T4** (leak) → euthyroid/hypothyroid phase → recovery · ★ **↓ RAIU**, ★ **↑ ESR**
- **Treatment:** mild → ★ **aspirin**; severe → ★ **prednisone 15–20 mg tid** + **propranolol** (thyrotoxic symptoms)

Subacute (de Quervain) thyroiditis: the three-phase course



Clue: after a viral URTI → PAINFUL tender thyroid (pain referred to jaw/ear/occiput), fever, dysphagia

Labs: ↑ T4 · ↓ RAIU · ↑ ESR

Treat: mild → aspirin · severe → prednisone 15–20 mg tid + propranolol for thyrotoxic symptoms

NOT: antithyroid drugs (nothing is being over-synthesised)

Added diagram: the three phases of subacute thyroiditis

B. ★★ Chronic lymphocytic (Hashimoto's) thyroiditis

- **Cause:** **autoimmune**; may coexist with **SLE**, **pernicious anaemia**, **Sjögren's**
- **Clinical:** goitre (may be asymptomatic) → **thyroid failure** usually follows
- Initially **T4 ↑**, then ↓ → **frank hypothyroidism**
- ★ **High-titre antimicrosomal (anti-TPO) antibodies**; can cause simultaneous thyroiditis + hyperthyroidism → ★ **hashitoxicosis**
- **Treatment:** ★ **levothyroxine (replacement doses)** → **goitre regresses**

	Subacute (de Quervain)	Hashimoto's
Cause	Viral , post-URTI	Autoimmune

	Subacute (de Quervain)	Hashimoto's
Gland	★ Painful, tender	★ Painless , firm goitre
ESR	★ ↑	Normal <i>[added]</i>
Antibodies	Usually absent <i>[added]</i>	★ High anti-TPO
RAIU (thyrotoxic phase)	↓	↓ (hashitoxicosis)
Outcome	Recovery (self-limited)	Permanent hypothyroidism
Treatment	Aspirin / prednisone + propranolol	Levothyroxine

📌 **Recap: Thyroiditis.** Subacute = painful post-viral gland, ↑ ESR, ↓ uptake, three phases; aspirin or prednisone. Hashimoto = painless autoimmune goitre, anti-TPO, ends in hypothyroidism; levothyroxine.

🔗 **Link:** Thyroiditis can feel nodular, but a single discrete lump raises a different question: is it cancer?

📌 **Up next: Solitary thyroid nodule.**

2.6 Solitary Thyroid Nodule (Slides 25-28)

- ★ **Very common:** clinically palpable in ~4%, found in ~50% at autopsy
- One prominent nodule on exam may be **multiple** on ultrasound
- ★ **Most are benign;** a small percentage are malignant; most thyroid cancers are **low-grade**
- ★ **Major aetiological factor for thyroid cancer: childhood/adolescent head & neck radiation**
 - Irradiated patients → **baseline US** + palpation **every 1-2 years**
- ★★ **FNA indications: dominant nodule > 1-1.5 cm** or **suspicious US features**
- ★ **Levothyroxine suppression of benign nodules is NO LONGER recommended** (nodules rarely shrink)

★★ **High-risk factors for malignancy in a nodule (Table 66-6):**

History	Examination	Lab & imaging
Head & neck irradiation	★ Hard consistency	★ ↑ Serum calcitonin (medullary)
Nuclear radiation exposure	★ Fixation	★ Cold nodule on technetium scan
Rapid growth	Lymphadenopathy	★ Solid lesion with microcalcifications on US
Recent onset	★ Vocal cord paralysis	
Young age	Distant metastasis	
★ Male sex		
Familial (medullary; ~5% of papillary)		

Table 66-6 High-Risk Factors for Malignancy in a Thyroid Nodule

History

Head and neck irradiation
Exposure to nuclear radiation
Rapid growth
Recent onset
Young age
Male sex
Familial incidence (medullary and about 5% of papillary)

Physical Examination

Hard consistency of nodule
Fixation of nodule
Lymphadenopathy
Vocal cord paralysis
Distant metastasis

Laboratory and Imaging

Elevated serum calcitonin
Cold nodule on technetium scan
Solid lesion with microcalcifications on ultrasonography

Slide 28: High-risk factors for malignancy in a thyroid nodule

🔄 **Recap: Nodule.** Common (4% palpable, 50% at autopsy), mostly benign. FNA if > 1–1.5 cm or suspicious on US. Red flags: radiation, male, young, rapid growth, hard/fixed, nodes, hoarseness, calcitonin, cold nodule, microcalcifications. Don't suppress with thyroxine.

🔗 **Link:** When the FNA shows malignancy, the cell type decides spread, treatment and prognosis.

▶ **Up next: Thyroid cancer types.**

2.7 Thyroid Cancer (Slides 29–31)

★★★ Cancer types (Slide 31 table):

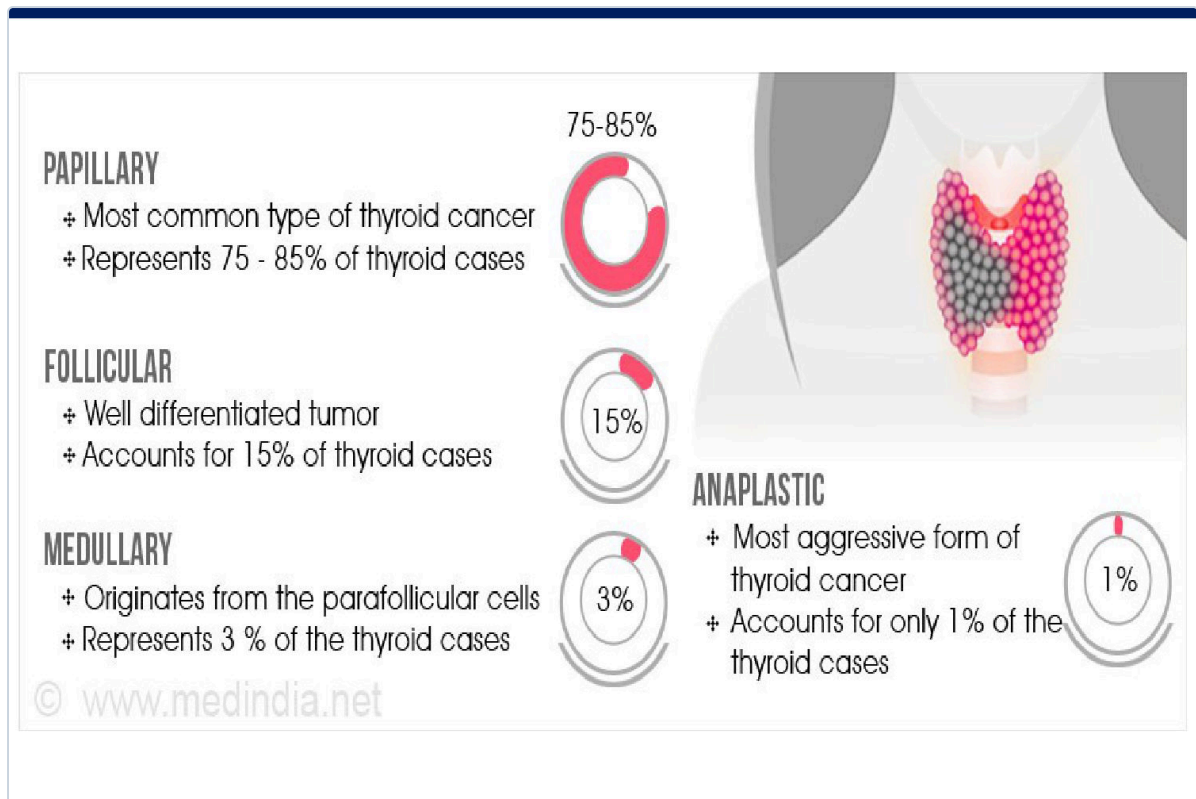
Type	Frequency	Behaviour	Spread	Treatment / prognosis
★ Papillary	70%	★ Commonest ; may be multifocal ; young people (F > M)	★ Lymph nodes ; local LN metastases even with occult tumours < 1 cm	Total thyroidectomy + adjuvant I-131 ; thyroxine for hypothyroidism and ↓ recurrence; ★ good prognosis
Follicular	20%	Single encapsulated lesion; mean age 50 y (F > M)	★ Blood → lung, bone	As papillary
★ Anaplastic	< 5%	★ Most aggressive ; hard, rapidly enlarging; weight loss, stridor , hoarseness	Locally invasive	Surgery/chemo/radio useless ; ★ 80% die within 1 year
Lymphoma	2%	Variable		Sometimes responsive to radiotherapy

Type	Frequency	Behaviour	Spread	Treatment / prognosis
★ Medullary	5%	From ★ calcitonin-secreting parafollicular C cells ; often familial , ★ MEN syndromes	Lymphatic spread frequent	★ Total thyroidectomy ; ★ I-131 has no role ; poor prognosis but indolent

Cell type	Frequency	Behavior	Spread	Treatment
Papillary	70%	Commonest. May be multifocal. Occurs in young people (F>M)	- Spreads via <u>lymph nodes</u> . - Local LN metastases occurs even with occult tumors < 1 cm.	Total thyroidectomy & adjuvant 131-iodine. Thyroxin give if hypothyroid and to reduce recurrence. <u>Good prognosis</u>
Follicular	20%	Single encapsulated lesion. Mean age 50 yrs. (F>M)	Spreads via <u>blood</u> , to lung/bone,...	As above
Anaplastic	<5%	<u>Most aggressive</u> . Hard rapidly enlarging. Wt loss, stridor, hoarseness	Locally invasive	Surgery, chemo, radio - <u>useless</u> 80% die within 1 yr.
Lymphoma	2%	Variable		Sometimes responsive to radiotherapy
Medullary cell	5%	Arises from calcitonin secreting parafollicular C cells. Often familial & associated with MEN syndrome.	Lymphatic spread has frequently developed.	Total thyroidectomy - 131-iodine has no role. - Poor prognosis, but indolent course

Slide 31: Thyroid cancer types table

Infographic (Slide 30): Papillary **75-85%** (most common) · Follicular **15%** (well-differentiated) · Medullary **3%** (parafollicular cells) · Anaplastic **1%** (most aggressive).



Slide 30: Thyroid cancer types infographic

Mnemonics [added]: "**Papillary = Popular, Palpable nodes, Positive prognosis**" · "**Follicular = Far (blood → bone/lung)**" · "**Medullary = MEN, calcitonin, No iodine**" · "**Anaplastic = Aggressive, Aged, Awful**". [added link to Lecture 1: thyroglobulin is the follow-up marker for papillary/follicular (differentiated) cancer; calcitonin is the marker for medullary.]

🔄 **Recap: Cancer.** Papillary (commonest, young, lymph nodes, excellent), follicular (blood → bone/lung), medullary (C cells, calcitonin, MEN, no I-131), anaplastic (elderly, fast, fatal within a year), lymphoma (radiotherapy). Differentiated types → total thyroidectomy + I-131 + thyroxine.

🔗 **Link:** The last clinical topic returns to function, specifically the grey zone where TSH is abnormal but T4/T3 are normal.

👉 **Up next: Subclinical hypo- and hyperthyroidism.**

2.8 Subclinical Thyroid Disease & Diagnostic Algorithms (Slides 32-40)

	Subclinical hypothyroidism	Subclinical hyperthyroidism
Definition	★ ↑ TSH, normal T4 & T3	★ ↓ TSH, normal T4 & T3
Symptoms	Usually none; subtle fatigue, constipation	Mild anxiety, palpitations, weight loss
Complications	↑ cardiovascular risk (esp. severe)	★ Atrial fibrillation , ★ bone loss , heart problems

★★ Managing subclinical hypothyroidism (Slide 34):

1. **Elevated TSH** → **confirm**; check **FT4, anti-TPO, lipids**
2. ★ **TSH ≥ 10 mU/L** → **treat with levothyroxine**
3. **TSH < 10**:
 - **Anti-TPO +** → **consider levothyroxine**

- **Anti-TPO – +** symptoms, goitre, young/middle-aged, **pregnant or planning pregnancy**, or dyslipidaemia → **consider levothyroxine**
- **Anti-TPO – +** **elderly** or none of the above → **close follow-up** (or treatment)

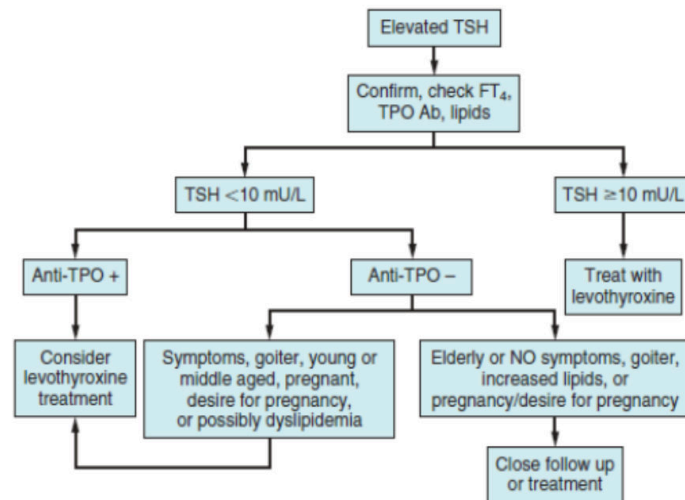


FIGURE 38-7 Management of subclinical hypothyroidism. Treatment is recommended for individuals with TSH 10 mU/L or greater. For individuals with TSH less than 10 mU/L, TPO antibody levels, age, pregnancy status, clinical symptoms/signs, and possibly lipids are important factors in considering whether to initiate treatment. FT₄, Free thyroxine; TSH, thyroid-stimulating hormone; TPO, thyroid peroxidase.

Slide 34: Management of subclinical hypothyroidism

★★ Managing subclinical hyperthyroidism (Slide 35):

TSH	> 60 y, heart disease, osteoporosis, or symptoms	< 60 y, no risk factors
★ < 0.1 mU/L (esp. if FT ₄ /T ₃ high-normal)	★ Treat	Observe or treat
0.1-0.5 mU/L	Clinical judgement (treatment suggested)	★ No treatment

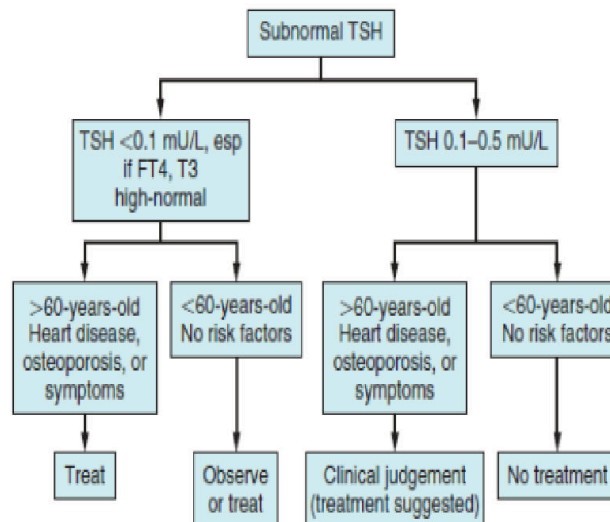


FIGURE 38-5 Management of subclinical hyperthyroidism. If the patient is older than 60 years or younger than 60 years with bone disease, heart disease, or symptoms, treatment is recommended. FT₄, Free thyroxine; TSH, thyroid-stimulating hormone.

Slide 35: Management of subclinical hyperthyroidism

★★★★ TSH-first testing (Slide 37):

TSH	FT4	Diagnosis
Low	Low	Central hypothyroidism or non-thyroidal illness (NTI)
Low	Normal	Subclinical hyperthyroidism or T3 toxicosis
Low	High	★ Thyrotoxicosis
Normal		Euthyroid
High	Low	★ Primary hypothyroidism
High	Normal	Subclinical hypothyroidism
High	High	★ TSH-secreting tumour
<i>TSH-first is for screening asymptomatic people; if symptoms are present, measure FT4 too.</i>		

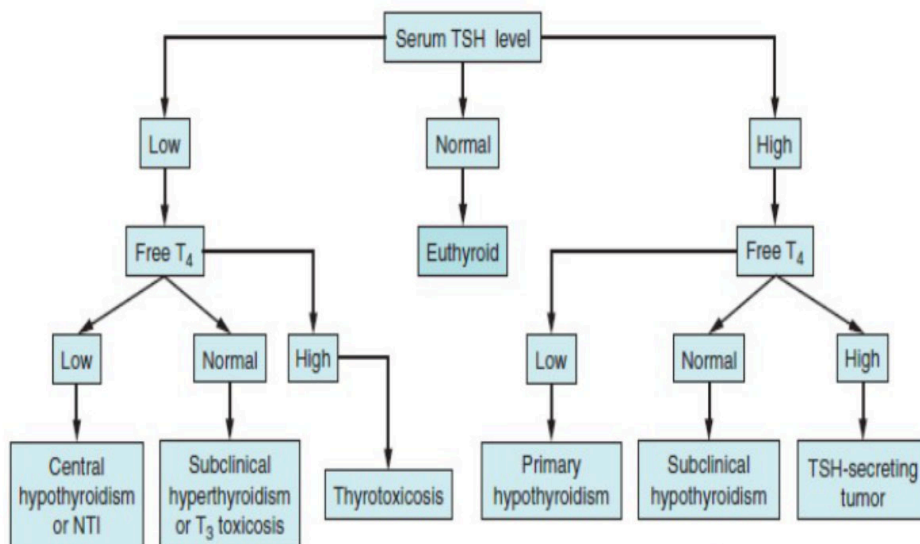
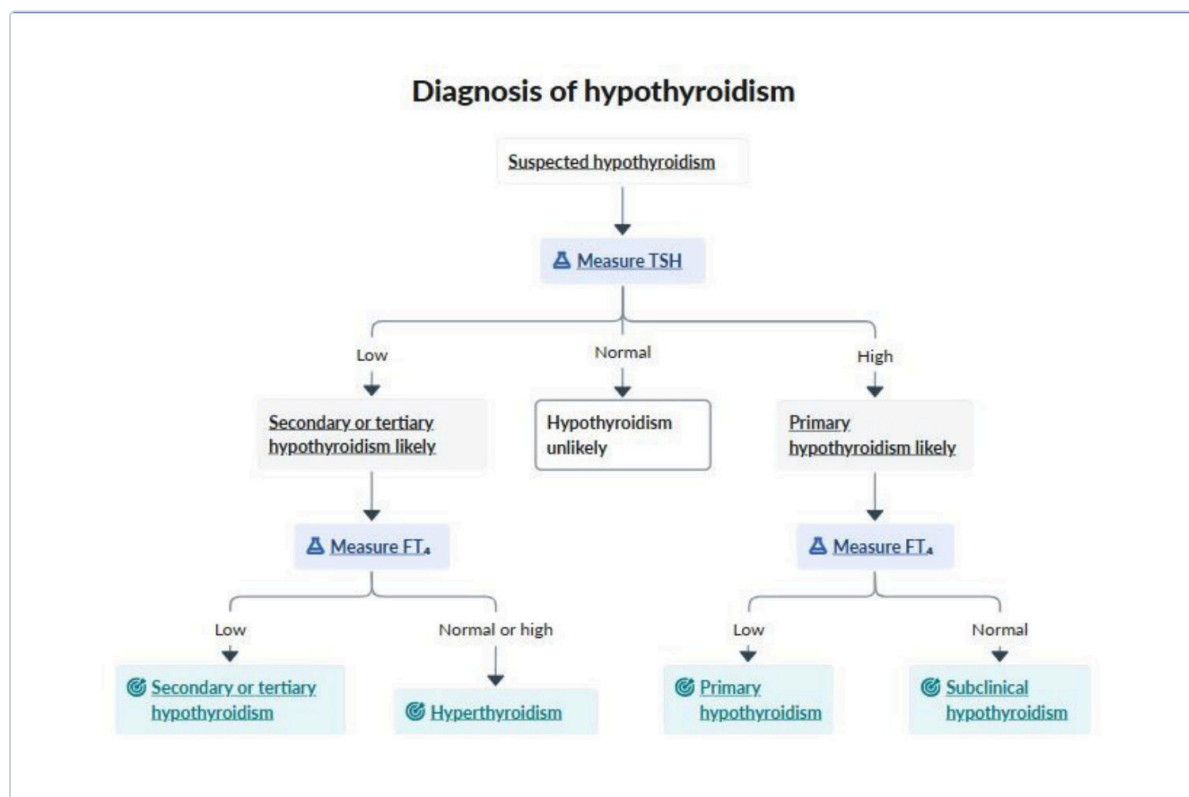


FIGURE 38-2 Sequential thyroid function testing using the serum thyroid-stimulating hormone (TSH) as a starting point. This strategy is useful for screening asymptomatic individuals. If symptoms are present, free thyroxine (FT₄) levels should also be assessed. NTI, Nonthyroidal illness; T₃, triiodothyronine.

Slide 37: Sequential thyroid function testing starting with TSH

★ **Diagnosis of hypothyroidism (Slide 39):** TSH → **low**: secondary/tertiary likely → FT₄ low = secondary/tertiary hypothyroidism (normal/high = hyperthyroidism) · **normal**: hypothyroidism unlikely · **high**: primary likely → FT₄ **low** = **primary**, **normal** = **subclinical**.

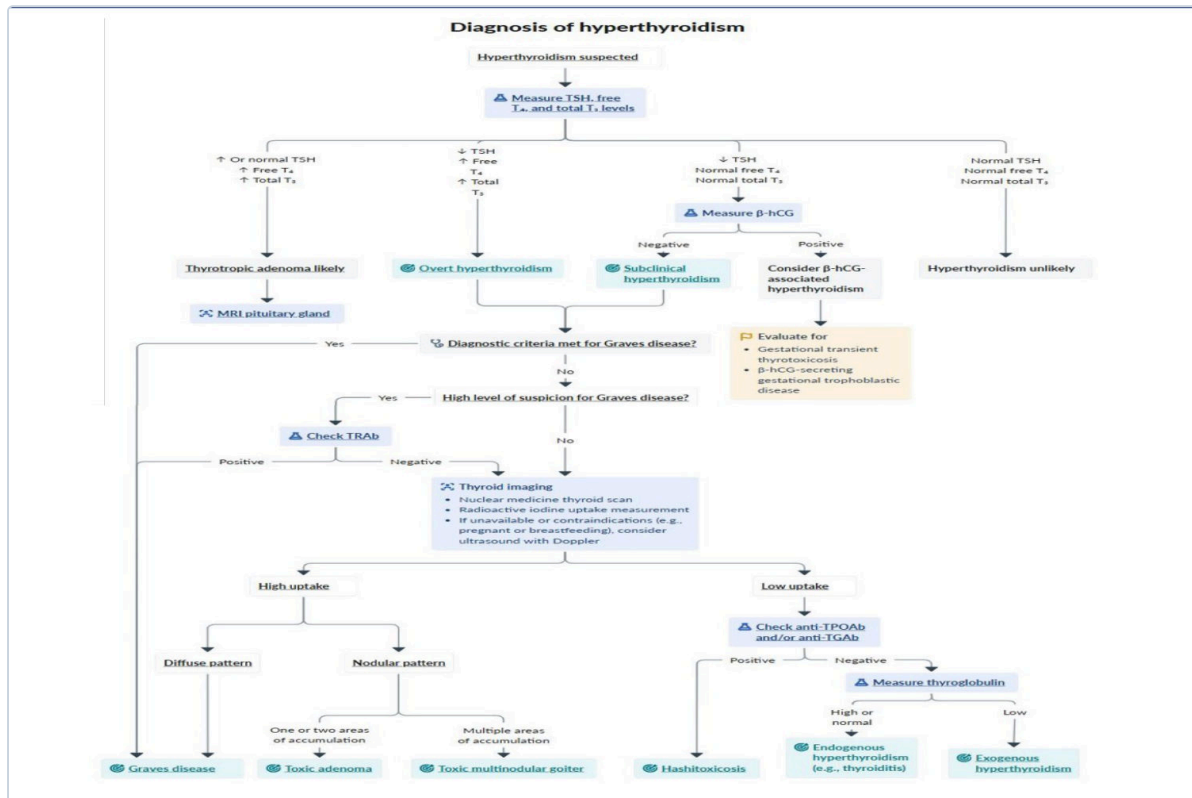


Slide 39: Diagnosis of hypothyroidism

★★ **Diagnosis of hyperthyroidism (Slide 38):**

1. Measure **TSH, free T4, total T3**

2. $\uparrow$ /normal TSH + $\uparrow$ FT4/T3 → ★ **thyrotropic (TSH) adenoma likely** → MRI pituitary
3. $\downarrow$ TSH + $\uparrow$ FT4/T3 → **overt hyperthyroidism**
4. $\downarrow$ TSH + **normal** FT4/T3 → measure ★ **β -hCG**: negative → **subclinical hyperthyroidism**;
positive → consider **hCG-associated** (gestational transient thyrotoxicosis, trophoblastic disease)
5. Overt/subclinical → **diagnostic criteria for Graves met?** Yes → **Graves**. If not, high suspicion → ★ **check TRAb** (positive → Graves)
6. Otherwise → **thyroid imaging** (nuclear scan / RAIU; **US with Doppler** if pregnant/breastfeeding):
 - **High uptake, diffuse** → **Graves**
 - **High uptake, nodular**: 1-2 areas → **toxic adenoma**; multiple → **toxic multinodular goitre**
 - **Low uptake** → anti-TPO/anti-Tg: **positive** → **hashitoxicosis**; negative → **thyroglobulin**:
high/normal → **endogenous (thyroiditis)**; ★ **low** → **exogenous hyperthyroidism**



Slide 38: Diagnosis of hyperthyroidism

★★ Diagnosis & management of thyrotoxicosis by aetiology (Slide 40, Table 38-2):

Aetiology	Mechanism	Gland	Key labs	RAIU	Treatment
Graves	Stimulating anti-TSH-R Ab	Diffuse goitre	+ antibodies, $\uparrow$ Tg	★ $\uparrow$ diffuse	ATDs, RAI, surgery
Toxic MNG / solitary nodule	Activating gene mutations	Nodule(s), usually > 3 cm	– TSI/TRAb, $\uparrow$ Tg	★ $\uparrow$ focal	★ RAI = treatment of choice ; ATDs temporising; surgery if compressive or cancer risk
Thyroiditis (thyrotoxic phase)	Follicular destruction	Tender (subacute) / non-tender ("silent")	– Abs, $\uparrow$ Tg	★ $\downarrow$	Prednisone
Iodine-induced	Iodine load (IV dye, amiodarone type 1)	Nodular or diffuse	– Abs, $\downarrow$ Tg	$\downarrow$	Withdraw agent; ATDs/RAI/surgery if selected

Aetiology	Mechanism	Gland	Key labs	RAIU	Treatment
Choriocarcinoma	hCG stimulates	Minimal goitre	↑ Tg	↑	Treat tumour
TSH-secreting pituitary tumour	TSH overproduction	Goitre	★ TSH normal/↑	↑	Treat adenoma
Struma ovarii	Ectopic tissue in ovarian tumour	Normal	↑ Tg	★ ↓ neck, ↑ pelvis	Treat tumour
Exogenous / factitious	Excess thyroxine intake	Normal	★ ↓ thyroglobulin	↓	Withdraw agent
Drug-induced	Amiodarone, interferon-α , tyrosine kinase inhibitors	Normal	↓ Tg	↓	Withdraw agent

Etiology	Mechanism	Thyroid Examination	Lab Results	Radioiodine Uptake	Treatment
Graves disease	Stimulatory anti-TSH receptor antibodies	Diffuse goiter	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ +Antithyroid Abs ↑ thyroglobulin	↑ uptake diffusely	ATDs RAI Surgery
Toxic multinodular goiter or solitary nodule	Activating gene mutations	Nodule(s), usually >3 cm	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ TSH/TRAb ↑ thyroglobulin	↑ focal uptake	RAI is the treatment of choice ATDs as temporizing measure Surgery if compressive symptoms or risk of thyroid cancer
Thyroiditis (thyrotoxic phase)	Follicular destruction causing release of stored hormone	Tender goiter in subacute thyroiditis; nontender goiter in "silent" thyroiditis	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ - Antithyroid Abs ↑ thyroglobulin	↓ uptake	Prednisone
Iodine-induced	Iodine surplus (IV dye or amiodarone-induced thyrotoxicosis type 1)	Nodular or diffuse goiter	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ - Antithyroid Abs ↓ thyroglobulin	↓ uptake	Withdrawal of offending agent if possible. ATDs, RAI, or surgery in selected cases
Choriocarcinoma	Secretes hCG, a thyroid stimulator	Minimal goiter	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ - Antithyroid Abs ↑ thyroglobulin	↑ uptake caused by increased thyroid hormone production	Treatment of choriocarcinoma
Pituitary tumor (TSH-secreting)	TSH overproduction	Goiter	TSH normal/↑ Normal/↑ FT ₄ Normal/↑ T ₃ - Antithyroid Abs ↑ thyroglobulin	↑ uptake	Treatment of pituitary adenoma
Struma ovarii	Ectopic thyroid tissue in ovarian tumor	Normal	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ - Antithyroid Abs ↑ thyroglobulin	↓ uptake in the region of the thyroid; ↑ uptake in the region of the pelvis	Treatment of struma ovarii
Exogenous thyrotoxicosis		Normal	TSH ↓ ↑ FT ₄ (if exogenous T ₄ intake) ↑ T ₃ (if exogenous T ₃ intake) - Antithyroid Abs ↓ thyroglobulin	↓ caused by suppression of thyroid hormone production	Withdraw offending agent
Iatrogenic or factitious	Over supplementation with thyroxine	Normal	TSH ↓ ↑ FT ₄ ↑ T ₃ - Antithyroid Abs ↓ thyroglobulin	↓ uptake	Withdraw offending agent
Drug-induced	Use of amiodarone, interferon-α, tyrosine kinase inhibitors	Normal	TSH ↓ ↑ FT ₄ ↑ T ₃ - Antithyroid Abs ↓ thyroglobulin	↓ uptake	Withdrawal of offending agent

Abs, Antibodies; ATDs, antithyroid drugs; IV, intravenous; RAI, radioactive iodine.

Slide 40: Diagnosis and management of thyrotoxicosis (Table 38-2)

🔄 **Recap: Subclinical & algorithms.** Subclinical = abnormal TSH with normal hormones. Treat subclinical hypo if TSH ≥ 10 (or < 10 with anti-TPO/symptoms/pregnancy). Treat subclinical hyper if TSH < 0.1 and > 60 y or heart/bone disease. Always start with TSH. In thyrotoxicosis, uptake + antibodies + thyroglobulin identify the cause.

🔗 **Link:** The closing case puts it together: a set of vague symptoms that point to one unifying diagnosis.

▶ **Up next:** Case, answered in §7.

3. High-Yield Comparison Tables

Table A: Hypothyroidism vs hyperthyroidism

	Hypothyroidism	Hyperthyroidism
Metabolic	Cold intolerance, weight gain, ↓ appetite	Heat intolerance, weight loss, ↑ appetite
Neuro	Fatigue, slow, delayed reflex relaxation	Irritable, brisk reflexes, fine tremor
Eyes	Periorbital oedema	Lid lag, exophthalmos (Graves)
CVS	Bradycardia , ↓ CO, diastolic HTN, effusion	Tachycardia, AF , systolic HTN
Skin	Cold, dry , orange (carotene)	Warm, moist
GI	Constipation	Hyperdefecation
MSK	Cramps, stiffness	Osteoporosis
Oedema	Generalised myxoedema	Pretibial myxoedema (Graves)
Shared	Proximal myopathy , menstrual disorders, ↓ libido/infertility, hair loss	
Lipids	↑ cholesterol	↓ cholesterol
Emergency	Myxoedema coma	Thyroid storm

Table B: Myxoedema coma vs thyroid storm

	Myxoedema coma	Thyroid storm
Trigger	Cold, infection, trauma, CNS depressants in untreated hypothyroid	Surgery, infection in untreated hyperthyroid
Temperature	★ Hypothermia	★ Hyperpyrexia
HR	Bradycardia	Tachycardia
Na / glucose	Hyponatraemia, hypoglycaemia	
Resp	Hypoventilation, CO₂ narcosis	
Steroid	Hydrocortisone 100 mg	Dexamethasone 4-8 mg
Hormone step	IV T4 500 µg	Methimazole/PTU → then iodide
Temperature Rx	Avoid heat loss (passive)	Ice bags

Table C: Subacute vs Hashimoto (see §2.5)

Table D: Thyroid cancer (see §2.7)

Table E: TSH / FT4 grid (see §2.8)

Table F: Subclinical disease: when to treat

	Treat	Consider / judgement	Observe
Subclinical hypo	TSH ≥ 10	TSH < 10 + anti-TPO+, symptoms, goitre, pregnancy (planned), dyslipidaemia	Elderly, TSH < 10, anti-TPO –
Subclinical hyper	TSH < 0.1 + > 60 y / heart / bone / symptoms	TSH 0.1-0.5 + risk factors; TSH < 0.1 + young	TSH 0.1-0.5, < 60 y, no risk

Table G: Thyroxine dosing

Situation	Regimen
Young, healthy	Full replacement (150–200 µg) <i>[added: ~1.6 µg/kg]</i>
Elderly / heart disease	25-50 µg, ↑ 25-50 µg monthly
Neonates, coma, emergency surgery	Rapid; IV + hydrocortisone
Myxoedema coma	500 µg IV
Monitoring	TSH at 6-8 weeks ; full effect 2-3 months

4. Rules, Exceptions, Patterns

- **95% of hypothyroidism is primary**; TSH ↑ is the key.
- **Nephrotic syndrome** lowers total T4 but **TSH is normal**; don't treat.
- **Rapid replacement only** for neonates, myxoedema coma and emergency surgery. **Everyone elderly/cardiac: start low, go slow.**
- **Give hydrocortisone with IV thyroxine** in coma or emergencies (occult adrenal insufficiency).
- **TSH lags**: check it at **6-8 weeks**, not sooner.
- **Synthetic T3 has no real indication.**
- **Subacute thyroiditis = painful + ↑ ESR + ↓ RAIU**; no antithyroid drugs.
- **Hashimoto's = painless, anti-TPO +**; treat with thyroxine; the goitre regresses.
- **FNA for nodules > 1-1.5 cm or suspicious US. No thyroxine suppression** for benign nodules.
- **Cold nodule and microcalcifications = suspicious.** Hot nodule is usually benign *[added]*.
- **I-131 treats papillary/follicular, not medullary or anaplastic.**
- **Papillary → lymph; follicular → blood.**
- **Subclinical hypo: treat at TSH ≥ 10. Subclinical hyper: treat at TSH < 0.1 with age > 60 or heart/bone disease.**
- **Low uptake + low thyroglobulin = exogenous thyroxine.**

5. Clinical Correlations

- **Carpal tunnel + constipation + bradycardia + fatigue + weight gain** → check TSH (lecture case).
- **Hyponatraemia, unexplained** → include hypothyroidism *[added: and adrenal insufficiency]* in the work-up.
- **Elderly hypothyroid patient with angina** → start 25 µg; rapid dosing can precipitate MI *[added]*.
- **Menorrhagia + microcytic anaemia** → consider hypothyroidism.
- **Postpartum woman with transient hyper → hypothyroid swing** → postpartum thyroiditis (anti-TPO > 60%).
- **Neck pain radiating to the jaw after a cold** → subacute thyroiditis (not dental or ENT pathology).
- **Hoarseness + hard fixed rapidly enlarging mass in an elderly patient** → anaplastic carcinoma.
- **Thyroid nodule + ↑ calcitonin + phaeochromocytoma / hyperparathyroidism** → **medullary cancer in MEN 2** *[MEN detail added]*.

- **Amiodarone** → can cause hypo- or hyperthyroidism [added]; type 1 (iodine-induced) is in the Slide 40 table.

6. Rapid Revision (Last-Day)

- Hypothyroidism: **95% primary**, 5% secondary; ♀ 30–50, > **80% antibody +**.
- Non-goitrous: congenital, **post-surgery, post-I-131**, idiopathic. Goitrous: **Pendred (+ deafness), iodine deficiency**, drugs (PAS, phenylbutazone, ATDs), **Hashimoto, Riedel**, subacute, postpartum.
- Myxoedema = mucopolysaccharide swelling (a sign).
- Neuro: **delayed reflex relaxation, carpal tunnel**, deafness, hoarseness, myxoedema madness.
- CVS: **bradycardia**, effusion, diastolic HTN, cardiomyopathy. Anaemia: **normo** (marrow), **megalo** (pernicious), **micro** (menorrhagia/achlorhydria).
- **Hyponatraemia, CO₂ retention, constipation/ileus**, ascites.
- Skin: **cold, dry, orange (carotene), outer 1/3 eyebrow loss**, xanthelasma. **Menorrhagia = commonest** reproductive sign.
- **Myxoedema coma**: hypothermia, hyponatraemia, hypoventilation, hypoglycaemia, hypotension. Triggers: **cold, infection, trauma, sedatives**.
- Rx coma: **T4 500 µg IV, hydrocortisone 100 mg, hypertonic saline + glucose, ventilation, avoid heat loss**.
- ECG: **low voltage + bradycardia**. Nephrotic: ↓ total T4, **normal TSH**.
- **Levothyroxine**; elderly **25–50 µg → +25–50 monthly → 150–200**. T4 normal in days, T3 **2–4 wk**, **TSH 6–8 wk**; full effect **2–3 mo**.
- Subacute: post-viral, **painful, ↑ ESR, ↓ RAIU**, 3 phases; **aspirin / prednisone 15–20 mg tid + propranolol**.
- Hashimoto: autoimmune (SLE, PA, Sjögren's), **anti-TPO high**, hashitoxicosis → **levothyroxine**.
- Nodules: **4% palpable, 50% autopsy**; radiation = main cancer risk; **FNA > 1–1.5 cm**; no T4 suppression.
- Red flags: radiation, rapid growth, young, **male**, family, **hard, fixed, nodes, cord palsy, calcitonin, cold nodule, microcalcifications**.
- Cancer: **papillary 70%** (lymph, young, good) · **follicular 20%** (blood → lung/bone, 50 y) · **medullary 5%** (C cells, MEN, **no I-131**) · **anaplastic < 5%** (80% dead in 1 y) · **lymphoma 2%** (radiotherapy).
- Subclinical hypo: ↑ TSH, normal T4 → **treat if TSH ≥ 10**. Subclinical hyper: ↓ TSH → AF, bone loss → **treat if < 0.1 + > 60 y/heart/bone**.
- TSH-first grid: ↑↑ = TSH tumour; ↓↓ = central/NTI.
- Hyperthyroid work-up: **β-hCG** if subclinical; **TRAb; RAIU** (diffuse/nodular/low); low uptake → antibodies → **thyroglobulin (low = exogenous)**.

7. Exam-Style Questions

Q1 (MCQ, lecture case, Slides 41–42). 50 ♀, 6 weeks of **right-hand pain and numbness (thumb/index), worse at night, relieved by shaking**. Episodic knee pain. Takes OTC medicine for **constipation**. **BMI 35. Fatigued, pulse 57**, BP 120/75. **DTRs 1+, mild leg oedema**. Mother has RA. Which treatment is most likely to benefit her? **A) L-thyroxine** B) Methotrexate C) Ibuprofen D) Surgical decompression E) Oral prednisone

Answer: A. [reasoning added] Carpal tunnel syndrome (mucinous infiltration of the flexor retinaculum) + constipation + bradycardia + fatigue + obesity + hyporeflexia + oedema =

hypothyroidism. Treating the cause with **levothyroxine** relieves the carpal tunnel. The family history of RA is a distractor (normal wrist range of motion, no synovitis), so methotrexate/prednisone are wrong. **Surgical decompression** is premature when the cause is reversible (no thenar atrophy).

Q2 (MCQ). A 78-year-old man with stable angina has TSH 45 mU/L and low FT4. Best initial levothyroxine dose? a) 200 µg daily b) 100 µg daily **c) 25 µg daily, increase monthly** d) 500 µg IV
Answer: c. Elderly/cardiac: **start 25-50 µg, ↑ 25-50 µg every month.** Full doses risk precipitating ischaemia. 500 µg IV is only for myxoedema coma.

Q3 (MCQ). An elderly hypothyroid woman is found drowsy after a winter night alone and a dose of sedative. T 33 °C, Na 121, glucose 3.0 mmol/L, PaCO₂ raised. Which is **wrong**? a) IV levothyroxine 500 µg b) IV hydrocortisone 100 mg c) Hypertonic saline + glucose **d) Active external rewarming with heating blankets** e) Ventilatory support **Answer: d.** The lecture says **avoid further heat loss** (passive measures). *[added: aggressive active rewarming causes peripheral vasodilatation and cardiovascular collapse.]*

Q4 (MCQ). A 45-year-old man has a 2-cm hard thyroid nodule; serum **calcitonin is raised**; his father had a pheochromocytoma. Which statement is correct? a) Radioiodine ablation is first-line b) It arises from follicular cells **c) Total thyroidectomy; I-131 has no role** d) It spreads mainly by the blood
Answer: c. Medullary carcinoma arises from **parafollicular C cells** (calcitonin), is often **familial / MEN**, spreads via **lymphatics**, and does not take up iodine.

Q5 (Short answer). A patient has TSH 7 mU/L with normal FT4. When would you treat? **Answer:** This is **subclinical hypothyroidism** with TSH < 10. Confirm, then check **anti-TPO and lipids**. **Consider levothyroxine** if **anti-TPO positive**, or if there are symptoms, goitre, young/middle age, **pregnancy or plans for pregnancy**, or dyslipidaemia. If elderly and none of these → **close follow-up**. Treat everyone at **TSH ≥ 10**.

8. Exam Pitfalls & Traps

1. **⚠ Cancer frequencies differ between slides:** table (Slide 31): papillary **70%**, follicular **20%**, medullary **5%**, anaplastic < **5%**, lymphoma **2%**. Infographic (Slide 30): papillary **75-85%**, follicular **15%**, medullary **3%**, anaplastic **1%**. The **rank order is the same**; know the order and quote the table if asked for exact numbers.
2. **Full-dose thyroxine in the elderly or cardiac patient.** Start low.
3. **Rechecking TSH at 2 weeks.** It takes **6-8 weeks** to settle.
4. **Forgetting hydrocortisone** in myxoedema coma or emergency surgery.
5. **Treating a nephrotic patient's low total T4.** TSH is normal; it's lost TBG.
6. **Actively rewarming** in myxoedema coma. Avoid heat loss instead.
7. **Antithyroid drugs for subacute thyroiditis.** Leak, not over-production.
8. **Confusing painful (subacute) with painless (Hashimoto, silent/postpartum) thyroiditis.**
9. **Thyroxine suppression for benign nodules** is outdated per the lecture.
10. **I-131 for medullary or anaplastic cancer.** No role.
11. **Papillary → blood, follicular → lymph.** It's the other way round: **papillary → lymph, follicular → blood.**
12. **Hot nodule = cancer.** It's the **cold** nodule that raises suspicion.
13. **Subclinical ≠ "no action".** Treat hypo at TSH ≥ 10; treat hyper at TSH < 0.1 in older/cardiac/osteoporotic patients.
14. **TSH ↑ and FT4 ↑ together** isn't lab error: think **TSH-secreting adenoma → MRI**.

15. **Low uptake thyrotoxicosis with low thyroglobulin** = factitious thyroxine, not thyroiditis.
16. **Hypertension in hypothyroidism** is real (↑ peripheral resistance). Don't assume BP must be low.
17. **Symbols lost in the PDF:** "much common in ___ than ___" on Slide 7 reads ♀ **than** ♂ on the rendered slide.